

Assignment One: Performance Modeling

SOFT 437: Software Performance Analysis

Winter 2026

Date Assigned: 23 January 2026

Date Due: 6 February 2026 at 11:59PM

Overview: The goal of this project is for you to demonstrate your ability to model the performance of an e-business website using the Use Case, Sequence Diagram, and Software Execution Model techniques we studied in class.

Background:

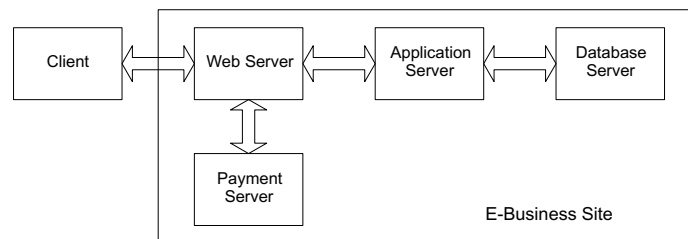


Figure 1: Servers for an E-Business Site

Figure 1 illustrates some of the common software servers that are used to support e-business functions. An e-business function is processed by several servers in an e-business site. Requests from browsers are clients of the services provided by the servers on the e-business site.

Figure 1 illustrates a typical logical configuration of an e-business site. The front-end server is a Web server. This server supports the HTTP protocol. It serves the home page of the site, processes requests to retrieve static HTML pages and launches programs (e.g., CGI scripts, servlets) that generate dynamic results. These programs are applications implemented by the application server. Examples of applications include searching for items in a catalog by keyword. In many cases, the application may need data stored in persistent storage (e.g., disk). Database management systems are used to provide efficient access to data in persistent storage. Thus, the application server may need to interact with the database server to obtain the data it needs to execute the requested operation. The result of the execution of an application is usually formatted as an HTML page that is returned by the Web server to the requesting client. Some of the functions provided by an e-business site require support for authentication. For example, an online trading site must be sure that a customer who logs into the site is providing accurate identification (authentication). A Payment server accepts payments online and transfers monetary funds from the customer's financial institution to the merchant's financial institution.

Consider an online bookstore using the architecture depicted in Figure 1. Customers can perform the following functions:

1. Connect to the home page and browse the site by following links to bestseller books and promotions of the week per book category.

2. Search for titles according to various criteria including keywords, author name, and ISBN.
3. Select one of the books results from a search and view additional information about the book such as a brief description, price, shipping time, ranking, and reviews.
4. Register as a new customer of the virtual bookstore. This allows the user to provide a username and a password, payment information, mailing address, and e-mail address for notification of order status and books of interest.
5. Login with a username and password.
6. Add items to the shopping cart.
7. Pay for the items added to the shopping cart.

Hints: You can check an existing e-commerce store, such as <https://www.amazon.ca/> to get familiar with the basic functions needed.

Miscellaneous Notes:

The Web Server, Application Server, Database Server and Payment Server are in the different computers. Moreover, the probability of requests for static pages to Web Server is 56%.

Based on the statistics, the average numbers of visits per function online bookstore are listed in Table 1.

Function	Visit Ratio (number requests per second)
Entry	1.000
Home	1.862
Browse	2.303
Search	2.193
Login	0.274
Pay	0.058
Register	0.196
Add to Cart	0.193
Select	0.919
Exit	1.000

Table 1: Average number of visits for each function

Deliverables:

1. Your group needs to prepare a report about this project. In particular, the report must include:
 - An introduction explaining SPE and performance modeling. The report should also present the reasons for using SPE to model the performance of this online bookstore.
 - Show the high-level structure of the online bookstore, which contains major functional components, their relationships on how the components interact with each other. Please also provide detailed design of the online bookstore which specifies the internal elements of all major system components, their properties, relationships and often their algorithms and the data structures. In your detailed design, please design the classes needed to implement a system

component and provide methods and data attributes contained within each class. Please provide description (e.g., the functionality, signature, parameters and return types) for each method and data attributes.

- Select all the critical use cases for performance and explain the criteria for your selection.
- You can create pick at most 3 critical use cases and create a sequence diagram for each of the identified critical use case. These diagrams should make use of references which can be expanded for more detail.
- A software execution model for each sequence diagram you created. You will need to add resource estimates for each node of your software execution model and solve the model. The processing overhead matrix is based on table 5-1 from the course's textbook.
- Create a deployment diagram to show the distribution of the components of the on-line bookstore of your design.
- A conclusion which explains your findings and recommendations for the corporation deploying the online bookstore. You should also discuss any troubling performance problems you uncovered through the model that you created.

Table 5-1: Web Server Computer Resource Requirements

Devices	CPU	Disk	Delay	GI/Net
Quantity	1	1	1	1
Service Units	K Instr.	I/Os	Visits	Msgs.
WorkUnits	25			
DB	500	4		
Msgs	25	1		1
Delay			1	
Service Time (sec.)	.000001	0.05	0.5	0.1

2. You need to need to deliver a presentation about this project for 10 minutes (including 7 minutes presentation for your report and 3 minutes for questions).
3. Submit the report in PDF to onQ by 11:59 PM on February 6th, 2026.
4. Submit the presentation in PDF to onQ by 2:00PM on February 9th, 2026.